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Monitoring and adapting endurance training on the basis of heart rate variability monitored by wearable technologies: A systematic review with meta-analysis

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Abstract

Background: Monitoring heart rate variability (HRV) as an indicator of daily variations in the functioning of the autonomic nervous system (ANS) may assist in individualizing endurance training to produce more pronounced physiological adaptations in performance.

Aims: To systematically perform a meta-analysis of the scientific literature to determine whether the outcomes of endurance training based on HRV are more favourable than those of predefined training.

Methods: PubMed, and Web of Science were searched systematically in March of 2020 using keywords related to endurance, the ANS, and training. To compare the outcomes of HRV-guided and predefined training, Hedges' g effect size and associated 95% confidence intervals were calculated.

Results: A total of 8 studies (198 participants) were identified encompassing 9 interventions involving a variety of approaches. Compared to predefined training, most HRV-guided interventions included fewer moderate- and/or high-intensity training sessions. Fixed effects meta-analysis revealed a significant medium-sized positive effect of HRV-guided training on submaximal physiological parameters ($g=0.296$, 95% CI 0.031 to 0.562, $p=0.028$), but its effects on performance ($g=0.079$, 95% CI -0.050 to 0.393 , $p=0.597$) and $\dot{V}O_2$ ($g=0.171$, 95% CI -0.213 to 0.371 , $p=0.130$) are small and not statistically significant. Moreover, with regard to performance, HRV-guided training is associated with fewer non-responders and more positive responders.

Conclusions: In comparison to predefined training, HRV-guided endurance training has a medium-sized effect on submaximal physiological parameters, but only a small and non-significant influence on performance and $\dot{V}O_{2peak}$. With regard to performance, there are fewer non-responders with HRV-based training.

KEYWORDS: cardiorespiratory fitness, endurance, mHealth, innovation, technology, training signature, wearable

Key Points

The effects of endurance training guided by heart rate variability (HRV) as an indicator of the functioning of the autonomic nervous system on parameters of performance are somewhat, although not statistically significantly better than those of predefined training.

With respect to performance, endurance training guided by HRV is associated with fewer who respond negatively and more who benefit.

In comparison to predefined training, HRV-guided training exerts a significant medium positive impact on submaximal physiological parameters.

Introduction

Changes in cardiorespiratory fitness and endurance performance in response to different predefined training programs **designed a priori to last for a specific period of time (e.g., 4 weeks)** vary widely – from positive, none or even negative adaptations [1-4]. Potential explanations for such variations include differences in age, sex, genetic predisposition and the type of training, as discussed in the scientific literature [2, 5-8]. Of these factors, only the type of training can be altered. Identifying the optimal approach to training can be challenging for coaches and/or athletes who are most often responsible for making this decision. Potentially they can be supported by the appropriate assessment and interpretation of autonomic cardiac regulation.

In this context, coaches may be aided by appropriate assessment and interpretation of autonomic cardiac regulation, which is known to play an important role in adaptations to endurance training [9]. In several different populations (e.g., athletes and obese individuals), adaptation of training to heart rate variability (HRV) mediated by the vagal nerve enhances improvements in peak ($\dot{V}O_{2peak}$) **and/or maximal ($\dot{V}O_{2max}$)** oxygen consumption and endurance performance (e.g., 5-km running time) [9-12]. Consequently, it has been proposed that endurance training based on daily alterations in HRV may reduce variation in physiological adaptations and potentially improve performance more effectively [9]. In the case of runners [13] and **other endurance athletes [14]**, recent systematic reviews comparing HRV-based and predefined training concluded that the former leads to more pronounced improvement in certain parameters related to performance, as well as in certain physiological variables (e.g., running velocity at the first or second lactate threshold) [13].

Technological developments, particularly in wearable sensor technology (wearables) and associated algorithms, have greatly facilitated the assessment and interpretation of HRV [3, 15]. These wearables are lightweight devices worn in, on, or close to the body and designed to monitor, analyse, and interpret a variety of parameters, as well as transmit and/or receive data from other devices and/or cloud services, often providing direct feedback to the user [16].

The aim of the present **systematic review with meta-analysis is to identify relevant scientific literature to compare the various effects of predefined training with training guided by variations in HRV with respect to performance or physiological parameters.**

Methods

Inclusion criteria

Our aim was to include all peer-reviewed publications that compare the outcomes (e.g., $\dot{V}O_{2\text{peak}}$ or $\dot{V}O_{2\text{max}}$) of predefined programs of endurance training with those based on variations in autonomic cardiac regulation.

Study populations

All investigations involving healthy endurance athletes, regardless of age, sex, gender, or level of performance, were included.

Interventions

To be eligible for consideration, the interventions were required to have the following characteristics, with appropriate reporting:

- Predominant involvement of endurance training
- Continuous adjustment of training by the intervention group on the basis of alterations in HRV
- Inclusion of a control group that performed a predefined program of training
- Utilization of wearable sensors
- Either null hypothesis significance testing (NHST) and/or magnitude-based inference (MBI) statistical analysis

With MBI statistical analyses, the potential benefit of the change observed is categorized as follows: most unlikely (<0.5%), very unlikely (0.5%–5%), unlikely (5%–25%), possible (25%–75%), likely (75%–95%), very likely (95%–99.5%), and most likely (>99.5%) [17]. When the 90% confidence limits overlap small positive or negative values, the benefit of the change is designated as unclear [17].

Outcomes

For inclusion, maximal indicators of performance, such as time-to-exhaustion during an appropriate test, in combination with peak or maximal (e.g., $\dot{V}O_{2\text{peak}}$ or $\dot{V}O_{2\text{max}}$) and submaximal (e.g., $\dot{V}O_{2\text{submax}}$, HR_{submax}) indicators of physiological condition had to be assessed.

Study Design

To be eligible for inclusion, the study had to be experimental (e.g., randomized controlled trials, pre/post design, and quasi-experimental investigations) and include a control group that performed predefined endurance training.

Publication status and language

We limited our search to full-length original articles published in English in peer-reviewed journals (i.e., so-called “grey” literature, such as conference abstracts, dissertations, theses or reports, was not included). In addition, the reference lists of articles initially included were examined for additional publications of potential relevance.

Search strategy

Following the recommendations of the Preferred Reporting Items of Systematic Reviews and Meta-analysis (PRISMA) [18], we carried out a systematic search of Ovid MEDLINE In-Process and Other Non-Indexed Citations, , and Web of Science (with no restriction concerning publication date) in March of 2020 to identify articles of potential relevance utilizing the following search profile: HRV OR heart rate variability OR autonomic nervous system AND guided OR training OR biofeedback AND endurance OR running OR exercise OR aerobic OR cycling OR swimming OR cross-country OR triathlon. **In addition, the list of references in each of the articles identified in this manner was screened in attempt to discover any other articles of relevance that might have been missed using this search profile.**

*****Figure 1 near here*****

Selection of articles

After incorporating potentially relevant articles into EndNote X9 (Clarivate Analytics, Philadelphia, PA, USA) and eliminating duplicates, one of us (CD) screened the titles and abstracts on the basis of our inclusion criteria and a second investigator (BS) independently confirmed those assessments. Thereafter, the full texts of the relevant articles were reviewed in detail for eligibility by these same two individuals (who were not blinded to the journals or authors involved). Cases in which there was disagreement were discussed, with or without input from a third author (CZ), until consensus was reached.

Data extraction and analysis

From each article thus identified, the first author extracted the details concerning publication (authors, journal, date); the participants (mean age, number, sex); the intervention (training mode, duration, wearables employed for monitoring HRV and related parameters, criteria for the modification of training); and the physiological and performance-related parameters evaluated (e.g., running performance, peak

oxygen uptake, running velocity at lactate thresholds). A second author (BS) confirmed that the data were extracted correctly.

Utilizing the commercially available Comprehensive Meta-Analysis software (CMA version 3.0, Biostat, Englewood, NJ, USA), meta-comparisons of the intervention and control groups before and after training, based on mean values and standard deviations, together with sample sizes and pre-post correlation values, were performed. When pre-post correlations were not reported, we employed the conservative estimate of $r=0.5$ recommended by Higgins and colleagues [19].

Hedges' g value was used as the measure of effect size (ES), with values of 0.8, 0.5, and ≤ 0.2 being considered large, medium and small, respectively. In addition, point estimates of the effect size and 95% confidence intervals were illustrated with forest plots. A positive ES indicated that the effect of HRV-based training was superior to that of pre-defined training.

Statistical heterogeneity was assessed with the Q [20] and I^2 [21] statistics. Due to the lack of substantial heterogeneity between studies, we applied a fixed-effects model for all comparisons [21].

Potential publication bias was explored using visual evaluation of potential funnel plot asymmetry, the Egger's linear regression test [22], the Begg and Mazumdar's rank correlation test (Kendall's S statistic $P-Q$) [23] and the Duval and Tweedie's trim-and-fill test [24].

The stability of the pooled ES for each individual study was assessed by sensitivity analysis approach, i.e., computation of the impact of removing that particular study from the analysis (leave-one-out). To help further ensure stable and reliable results, a cumulative meta-analysis was conducted. For all analysis, $p<0.05$ was considered to be statistically significant.

Assessment of methodological quality

The quality of the methodology employed in each of the articles included was assessed using the scale of the Physiotherapy Evidence Database (PEDro) (described in detail elsewhere [25, 26]). In brief, ten

different aspects of each intervention (e.g., blinding of subjects and assessors, the extent to which the subjects in the experimental and control groups differed prior to the intervention) were assessed and one point assigned for each positive assessment, resulting in a highest possible score of 10 points [25, 26]. Two of the authors (PD and CZ) assessed the risk of bias independently, resolving any disagreements through discussion, with or without the involvement of a third author (BS).

Results

Study characteristics

The total number of records identified and examined are shown in Figure 1. **5169 article were identified in Web of Science, 5652 in Pubmed.** Of these articles, eight met our inclusion criteria for analysis. All of these were published between 2007 and 2019 and received PEDro scores of 6-8 (exact PEDro score of each article can be found in the appendix).

Table 1 presents key aspects of the interventions analysed, including the redefined program of training, methodology used to monitor HRV, and alteration of training by the intervention group in response to changes in HRV.

***** Table 1 not here*****

In one case, two intervention groups were involved (Kiviniemi et al. 2010), i.e., the 8 studies involved a total of 9 interventions.

Altogether, 198 participants (55 women, 128 men and 15 participants in one study whose sex was not reported) performed training interventions ranging from 15 days [27] to four [5] or eight weeks [9, 28-32] in duration. The $\dot{V}O_{2peak}$ values for the male and female participants ranged from 49-68 ml·kg⁻¹·min⁻¹ [5, 9, 29-32], and 35-59 ml·kg⁻¹·min⁻¹ [27-29], respectively. (One article [28] did not report this value prior to the intervention). Three interventions consisted of running only [5, 9, 28]; another involved running approximately 60% of the training time (with ~13% cycling and ~27% other) [29]; one running in combination with five sessions of strength training during the 8-week intervention [30]; two cycling [32]; and one hypoxic “live-high-train-low” Nordic skiing [27].

In connection with 7 of the interventions [5, 9, 27-30], null-hypothesis testing alone was applied to assess intra- and/or intergroup differences, while the remaining two also used magnitude-based inference

statistics [31, 32].

During 8 interventions HRV was assessed in the morning [5, 9, 27, 29-32] and in the afternoon/evening in the remaining intervention [28]. The parameters related to HRV that were assessed varied: in association with five interventions the root mean sum of squared differences (rMSSD) was determined [9, 28, 30-32]; one focused on high-frequency power [5]; one the standard deviation associated with the instantaneous beat-to-beat variability in the R-R interval (obtained from Poincaré plots) [29]; and the other spectral power analysis [27, 29].

The periods used for determination of the baseline HRV for comparison with the daily values observed during the intervention were one [27], three [30], seven [9, 31-32] and 10 days [5, 28, 29].

Training parameters

In connection with 5 of the interventions, modifications in training based on alterations in HRV resulted in fewer moderate- and/or high-intensity sessions of exercise than with a predefined program of training [5, 9, 28, 29, 31]; a greater number of moderate-intensity sessions in one intervention [29]; and no difference in three cases [27, 30, 32] (Table 2).

***** Table 2 near here*****

Parameters used as indicators of performance

Performance was assessed on the basis of peak power output, the results of a 40-min time-trial, time required to run 3000 m and/or 5000 m, maximal running velocity, and time required to attain this velocity in 7 of the studies [5, 9, 28-32], which we have grouped together here. Table 2 summarizes the pre- and post-values for these parameters.

The 14 ESs in combination demonstrated a small positive, but statistically insignificant effect (ES=0.171, SE=0.113, 95% CI -0.050 to 0.393, Z-value=1.514, p=0.130; Figure 2) of HRV-guided training on

performance parameters. No statistically significant heterogeneity was detected ($Q=8.918$, $df=13$, $p=0.779$; $I^2=0.000\%$).

*****Figure 2 near here*****

Publication bias

Visual inspection of the funnel plot (Figure 3) showed no evidence of publication bias concerning parameters related to performance, a conclusion confirmed by Begg and Mazumdar's rank correlation test (Kendall's S statistic $P-Q=19.00$; tau without continuity correction= 0.209 , $z=1.04$, $p=0.149$; tau with continuity correction= 0.198 , $z=0.985$, $p=0.162$), as well as by Egger's linear regression test (intercept= 0.735 , $SE=1.706$, 95% CI -2.981 to 4.552 , $t=-0.431$, $df=12$, $p=0.337$). The Duval and Tweedie's trim-and-fill test did not identify any missing study on either side of the plot.

*****Figure 3 near here*****

Sensitivity and cumulative meta-analysis

The reliability and stability of our findings on parameters related to performance were confirmed by the sensitivity and cumulative meta-analyses (illustrated in the relevant supplementary figure).

Peak oxygen uptake

In 6 of the studies [5, 27, 29-32], $\dot{V}O_{2peak}$ was assessed (Table 3) summarizes the values obtained before and after the interventions.

*****Table 3 near here*****

The 9 ESs in combination revealed a small-positive, but statistically insignificant effect ($ES=0.079$,

SE=0.149, 95% CI -0.213 to 0.371, Z-value=0.529, $p=0.597$; Figure 2) of HRV-guided training on $\dot{V}O_{2peak}$, with no significant heterogeneity ($Q=3.227$, $df=8$, $p=0.919$; $I^2=0.000\%$).

Publication bias

A visual inspection of the funnel plot (Figure 3) showed no evidence of publication bias regarding $\dot{V}O_{2peak}$; a conclusion confirmed by Begg and Mazumdar's rank correlation test (Kendall's S statistic $P=Q=14.00$; tau without continuity correction=3.389, $z=1.46$, $p=0.07$; tau with continuity correction=-0.361, $z=1.355$, $p=0.088$), as well as by Egger's linear regression test (intercept=2.543, SE= 1.695, 95% CI -1.464 to 6.551, $t=1.501$, $df=7$, $p=0.09$). The Duval and Tweedie's trim-and-fill test did not identify any missing study.

Sensitivity and cumulative meta-analyses

In summary, the reliability and stability of our findings concerning $\dot{V}O_{2peak}$ were confirmed by the sensitivity and cumulative meta-analyses (illustrated in the relevant supplementary figure).

Submaximal physiological parameters

For this meta-analysis, the data from the 6 studies [5, 9, 27, 30-32] (comprising 61 participants) that evaluated the effects of both HRV-guided and predefined training on submaximal physiological parameters before and after the intervention were pooled (Table 3). The effects on lactate thresholds 1 (LT1) and 2 (LT2) were evaluated by Vesterinen (2016), Nuuttila (2016) and co-workers; while Javaloyes et al. (2018, 2019) evaluated the effects on ventilatory thresholds and the corresponding power output (i.e., WVT1, WVT2). Each parameter evaluated was analysed independently.

The 10 ESs in combination demonstrated a statistically significant small-to-medium positive effect ($ES=0.296$, $SE=0.135$, 95% CI 0.031 to 0.562, $Z\text{-value}=2.191$, $p=0.028$; Figure 2) of HRV-guided training on submaximal physiological parameters. The heterogeneity calculated ($Q=6.424$, $df=9$, $p=0.697$; $I^2=0.000\%$) was insignificant.

Publication bias

Visual inspection of the funnel plot (Figure 3), as well as the results of the Begg and Mazumdar's test (Kendall's S statistic $P-Q=37.00$; tau without continuity correction=0.822, $z=3.309$, $p=0.000$; tau with continuity correction=0.8, $z=3.219$, $p=0.000$) and Egger's linear regression test (intercept=4.379, $SE=1.291$, 95% CI 1.402 to 7.357, $t=3.391$, $df=8$, $p=0.005$) indicated publication bias with respect to the submaximal physiological parameters. Correction of this asymmetry using Duval and Tweedie "trim and fill" method yielded 4 potentially missing studies, and an adjusted ES of 0.458 (95% CI 0.241 to 0.675, $Q=10.808$).

Sensitivity and cumulative meta-analyses

The sensitivity and cumulative meta-analyses confirmed the reliability and stability of these findings on submaximal physiological parameters (illustrated in the relevant supplementary figure).

Variations in performance

Four of the articles described the response to HRV-based and predefined training with respect to performance during a 40-minute time-trial and 3000-m running, as well as maximal running velocity [9, 30-32].

In two cases fewer negative responders were present in the intervention group [31, 32]. Indeed, of the 16 participants in these two interventions, only two demonstrated a decline in performance (by approximately 1% and 20.5%) during the 40-minute time-trial (40TT). In contrast, 5 of the 16 participants in the control

groups that performed predefined training exhibited poorer 40TT performance (with declines of -2%, -5% [32], and -11.6%, -9.1% and -2.7% [31]. In both of these studies, the average improvement in 40TT performance was more pronounced following HRV-based than predefined training [31, 32]. Another study found that the variation in 3000-m running performance was smaller following HRV-based (-1% to 6%) than predefined training (-4% to 8%) [9]. In three other studies, HRV-based training led to the greatest individual improvements in performance (i.e. in 40TT, maximal running velocity) [30-32].

Discussion

This systematic review and meta-analysis compared the effects of predefined training with those of training guided by variations in HRV, as detected employing wearables.

The major observations were as follows:

- 1) 5 of the 9 interventions involved fewer sessions of moderate- and/or high-intensity exercise than predefined training.
- 2) In comparison to predefined training, HRV-guided training exerts a small positive, but statistically insignificant effect on parameters related to performance (i.e., peak power output, performance in connection with a 40-min time-trial or 3000 m and/or 5000 m of running, maximal velocity, and time required to attain this velocity) and $\dot{V}O_{2peak}$.
- 3) In comparison to predefined training, HRV-guided training had a significant moderate-positive effect on submaximal physiological parameters.
- 4) **With respect to performance, some participants demonstrated worsening, but more showed improvement following HRV-guided training.**

- 5) Publications in this area to date have involved a wide variety of methodological approaches (e.g., different parameters related to HR, different training interventions). Similar research on, e.g., a variety of different study populations is desirable.

Evaluation of the functional status of the autonomic nervous system (ANS) could reveal important aspects of physiological responses to training [33, 34], since this system is the primary regulator of fundamental aspects of overall recovery, including delivery of oxygen and nutrients, removal of waste products, and thermoregulation by the cardiovascular system [35].

As described eloquently elsewhere [5], extensive vagal activity prior to training for 8 weeks (six 30-40 min sessions per week at 70-80% peak heart rate) improves aerobic power [11]. On the basis of other previous findings it has been proposed that the more pronounced enhancement in average performance following HRV-guided endurance training is due to more optimal timing of (more intensive) training sessions [5, 31, 32].

In this connection our present meta-analysis indicates the primary involvement of improvements in submaximal physiological variables (moderate effect), with smaller improvements in $\dot{V}O_{2peak}$ (small effect) exerting less impact. However, additional investigations on the physiological mechanism(s) underlying these improvements in performance clearly need to be carried out, especially since the methodology employed in the articles reviewed here varied considerably.

The only 4 studies [9, 30-32] that evaluated individual variations in the effects of training on performance (i.e., during a 40-minute time-trial or 3000-m running, as well as maximal running velocity) demonstrated that in contrast to predefined training, HRV-guided training resulted in fewer participants with poorer performance, whereas a larger number showed more extensive improvement. Thus, for the individual HRV-guided training appears to produce results that are more favourable than those obtained with predefined training.

Of course, at present this proposal is highly preliminary **and a more detailed understanding of the processes involved is required. Studies analysed changes in performance at the individual level were simply counted and/or reported, but such “responder” counts are biased due to random intraindividual variability in the outcome(s) assessed before and after the intervention [35]. New approaches for estimating the proportion of responders in a population of interest that are less prone to bias involve the standard deviation for true individual responses, thereby directly incorporating information about the control group [35]. In addition, none** of the investigations examined interindividual variations in the effects of training on physiological parameters (e.g., $\dot{V}O_{2peak}$, submaximal parameters) and such information could help elucidate the mechanisms underlying the more pronounced effects of HRV-guided training on performance.

With the growing availability of novel sensors and algorithms [36] that effectively monitor heart rate, it is becoming more and more feasible to base one's training on variability in this parameter. Recreational and elite athletes who perform endurance training without a coach may benefit most in this respect [15]. However, most of the end consumer oriented wearable sensors designed to monitor HRV have yet to be validated and consumers must always keep this limitation in mind [36].

The strengths of this systematic review and meta-analysis include our comprehensive search of the literature (including more than two scholarly databases, as recommended by the Cochrane Association guidelines and good practices for conducting systematic reviews), followed by careful appraisal of the quality of the studies included. Interestingly, the methodological approach in all of the publications we review here was of high quality, as indicated by their PEDro scores. In addition, there was no evidence for any significant heterogeneity across these studies, which further strengthens our conclusions.

With respect to the parameters related to performance (peak power output, during a 40-min time-trial or 3000-m and/or 5000-m run, maximal velocity, and time required to attain that velocity) and $\dot{V}O_{2peak}$, there was no evidence of publication bias. However, this was not the case for submaximal physiological

parameters and in this connection our results must be interpreted with caution.

We did not contact researchers in sports and exercise sciences for information on ongoing studies or manuscripts in press or available as pre-prints **and our search was limited to full-length original articles in English and published in peer-reviewed journals**, which means that we might have missed some important findings.

In four of the publications [27, 30-32], the sample size required to reach significant conclusions was not calculated *a priori* (e.g., using G*Power software [37]). Many other aspects of the methodologies involved (e.g., **wearable sensors** and the procedures utilized to measure different parameters related to HRV, alteration of training in response to variations in this parameter) varied **between studies, which complicates comparison. This variation, along with the current availability of only relatively few studies, makes** determination of a “best-practice” approach impossible at present. However, it has been argued previously that derivation of rMSSD and SD1 from a Poincaré plot is the most useful approach for close monitoring of athletes [38].

It remains unclear how training should best be altered in response to a reduction in HRV. For instance, should training be paused completely or high-intensity training replaced with low-intensity sessions? In addition, none of the articles included strategies for recovery, e.g., immersion in cold water, massages or other measures designed to reduce fatigue following endurance exercise [39]. It is possible that inclusion of such procedures would be more beneficial to performance and/or physiological variables than alterations in training.

In addition, numerous factors other than training, including psychological stress related to work or academic studies, exert an impact on HRV [33] and, therefore, we cannot be entirely certain what caused changes in the HRV in connection with the interventions described here. In this context, characterization of individual stress [3] would be desirable, potentially leading to strategies for reducing this stress and

thereby improving recovery and/or training.

Conclusions

In comparison to predefined endurance training, such training that is adapted in response to changes in HRV has a significant medium effect on submaximal physiological parameters, at the same time as its effects on performance and $\dot{V}O_{2peak}$ are small and statistically non-significant. However, with respect to performance, HRV-based training tends to be associated with fewer negative and a larger proportion of positive responders. However, the scientific literature in this area is sparse and the studies that have been performed have employed a variety of methodological approaches, rendering definitive conclusions impossible to draw at the present time.

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Conflict of Interest:

PD, CZ, KT, JR, HC, PK and BS have no conflicts of interest to declare.

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Table_appendix

The PEDro scores for the articles included in the present analysis

Article	Item on the PEDro scale											Total
	1 ^a	2	3	4	5	6	7	8	9	10	11	
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007)	1	1	1	1	0	0	0	1	1	1	1	7
(Kiviniemi et al., 2010)	1	1	1	1	0	0	0	1	1	1	1	7
(Vesterinen et al., 2016)	1	1	1	1	0	0	0	1	1	1	1	7
(da Silva, Ferraro, Adamo, & Machado, 2017)	1	1	1	1	0	0	0	1	1	1	1	7
(Nuutila, Nikander, Polomoshnov, Laukkanen,	1	1	1	1	0	0	0	1	1	1	1	7

& Hakkinen, 2017)													
(Schmitt, Willis, Fardel, Coulmy, & Millet, 2018)	1	1	1	1	0	0	0	1	1	1	1	1	8
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018)	1	1	1	1	0	0	0	1	1	1	1	1	8
(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019)	1	0	1	1	0	0	0	0	1	1	1	1	6

^a not included in calculation of the total PEDro score

- da Silva, D. F., Ferraro, Z. M., Adamo, K. B., & Machado, F. A. (2017). Endurance Running Training Individually-Guided By Hrv In Untrained Women. *J Strength Cond Res*. doi:10.1519/JSC.0000000000002001
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- Schmitt, L., Willis, S. J., Fardel, A., Coulmy, N., & Millet, G. P. (2018). Live high-train low guided by daily heart rate variability in elite Nordic-skiers. *Eur J Appl Physiol*, 118(2), 419-428. doi:10.1007/s00421-017-3784-9
- Vesterinen, V., Nummela, A., Heikura, I., Laine, T., Hynynen, E., Botella, J., & Hakkinen, K. (2016). Individual Endurance Training Prescription with Heart Rate Variability. *Med Sci Sports Exerc*, 48(7), 1347-1354. doi:10.1249/mss.0000000000000910

Confirmation of Ethical Compliance

Not Applicable

Conflict of Interest:

PD, CZ, KT, JR, HC, PK and BS have no conflicts of interest to declare.

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Table 1: Characterization of the studies analysed

Article	Participants		Sport	Predefined training characteristics	HRV Guided Training		
	N (male, female)	Age (yrs)			Technology employed for monitoring HRV	Measurement employed to adapt training	Modification of training
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018)	17 (17,0)	39.2 ± 5.3 (HRV-based) 37.62 ± 7.09 (Predefined)	Cycling	8-week intervention period Maintenance of total training volume Increased in training intensity (not specified any further) 4 x (30 min at VT2) 4 x (4 x 8 >VT2) 6 x (40 min between VT1 & 2) 20-32 x (120 min < VT1)	Polar H7 Chest strap (Polar Electro OY, Kempele, Finland)	rMSSD determined every morning 90 sec in supine position (first 30 sec discarded) 7d moving window	No more than 2 consecutive days of rest No more than 2 consecutive sessions of moderate- or high-intensity training If rMSSD was less than the smallest worthwhile change, training intensity changed from moderate or high-to-low or rest
(Javaloyes, Sarabia, Lamberts,	15 (n.r.)	28.1 ± 13.2 (HRV-based)	Cycling	8-week intervention period	Smartphone flashlight and camera were used to assess pulse-rate	rMSSD was determined every morning	No more than 2 consecutive

Plews, & Moya-Ramon, 2019)		30.8 ± 10.5 (Predefined)	(HRV-based) 59.0 ± 6.2 ml·kg ⁻¹ ·min ⁻¹ (Predefined)		Maintenance of total training volume 4 training blocks: 3 high-intensity (consisting of 4 sessions (≥VT2) and HIITs (>VT2) per week) followed by a block of low-intensity (<VT1) training (4 sessions and 1 HIIT per week)	variability at the fingertip using the smartphone application HRV4Training (www.hrv4training.com)	90 sec in supine position 7d moving window of was calculated to guide training	days of rest No more than 2 consecutive sessions of high-intensity training If rMSSD was below the smallest worthwhile change, training intensity changed from high to low or rest.
(Schmitt, Willis, Fardel, Coulmy, & Millet, 2018)	24 (19, 5)	23.3 ± 3.6 (male) 22.8 ± 4.1 (female)	$\dot{V}O_2$ max: 67.8 ± 3.6 ml·kg ⁻¹ ·min ⁻¹ (male) 58.9 ± 2.5 ml·kg ⁻¹ ·min ⁻¹ (female)	Nordic skiing	HRV- based: 15 days of “live-high-train-low” sleeping with 15% O ₂ in the air Hypoxic predefined: sleeping with 15% O ₂ in the air Normoxic predefined All three groups trained in a similar manner as recommended by the national coach Training intensities 1) < VT1 2) VT 1 – VT2 3) At VT2 4) For strength and speed sessions	Ambic 2 Peak (Suunto Oy, Vantaa, Finland)	HRV analysed every morning RR intervals between the 3 rd and 8 th min supine, and between the 9 th and 14 th min standing. Spectral Power analysis with fast Fourier transformation (high-frequency power band 0.15-0.40 Hz; low-frequency power band 0.04-0.15 Hz)	Maintained or increased HF induced a higher TL than on the previous day. A ≥30% decrease in HF led to lower TL. Two consecutive days of reduced HF led to a day of rest. Following a day of rest, a session of low-intensity and medium-volume exercise was prescribed.
(Vesterinen et al., 2016)	20 (20, 0)	34 ± 8	$\dot{V}O_2$ peak: 49 ± 4 ml·kg ⁻¹ ·min ⁻¹	Running	38 training sessions 8 weeks 16 at low-	Omegawave Pro Mobile System (Omegawave Ltd., Helsinki, Finland)	rMSSD determined every morning 4-min supine position	4-week baseline period immediately before the intervention

				intensity (below LT1) 16 at moderate or high-intensity Periodization: 3 weeks of intense training, 1 week of easy training, progressively increasing intensity 1 session of activity other than running each week		7d moving window of	Smallest individual worthwhile changes calculated from the baseline data If 7d moving-window rMSSD < smallest worthwhile change: low-intensity training. If 7d moving-window rMSSD > smallest worthwhile change: high- or moderate-intensity training
(Kiviniemi et al., 2010)	28 (14, 14)	Men: 35 ± 4 Women 33 ± 4	$\dot{V}O_{2peak}$: Men: 50 ± 6 ml·kg ⁻¹ ·min ⁻¹ Women: 35 ± 5 ml·kg ⁻¹ ·min ⁻¹	Running 2 weeks training intervention 2x40 min at ~70% HR _{max} 3x30 min at ~85% HR _{max} (+5 min warm-up and cool-down)	Polar RS800 (Polar Electro OY, Kempele, Finland)	SD1 determined every morning 3 min standing	Individual baseline data from the preceding 10 days (moving window) If no significant decrease in SD1 on any given day: vigorous exercise Moderate exercise after 2 sessions of vigorous exercise Significant decrease in SD1 on any given day: moderate intensity exercise Significant decrease in SD1 on 2 consecutive days: rest

								for at most 2 consecutive days, followed by moderate-intensity exercise Moderate exercise after 2 sessions of vigorous exercise
(Kiviniemi et al., 2010)	10 (0, 10)	35 ± 4	$\dot{V}O_{2peak}$: 37 ± 5 ml·kg ⁻¹ ·min ⁻¹	Running	8 weeks training intervention 2x40 min at ~70% HR _{max} 3x30 min at ~85% HR _{max} (+5 min warm-up and cool-down)	Polar RS800 (Polar Electro OY, Kempele, Finland)	SD1 determined every morning Every morning 3-min standing	Individual baseline data from the preceding 10 days (moving window) Vigorous exercise only when HRV increased Moderate intensity exercise when HRV remained unchanged Moderate exercise after 2 sessions of vigorous exercise
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007)	26 (26, 0)	35 ± 4	$\dot{V}O_{2peak}$: ~56 ± 4 ml·kg ⁻¹ ·min ⁻¹	Running	4 weeks training intervention period Day 1: Low-intensity exercise; Day 2: High-intensity; Day 3: High-intensity; Days 4-6: Repeat Days 1-3; Day 7: Rest High-intensity: 30 min at 85% HR _{max} (+5 min warm-up and cool-down at 65% HR _{max}) Low-intensity: 40 min at 65% HR _{max}	Polar S810i (Polar Electro OY, Kempele, Finland)	High-frequency power analysis (HF _{power} ; 0.15-0.40 Hz) of the spectral component of HRV determined every morning 5-min standing	Individual baseline data collected on the preceding 10 days (moving window) If HF _{power} increased significantly on any given day: 2 days of high-intensity exercise followed by low-intensity exercise If HR _{power}

								decreased significantly on any given day: low-intensity training If HF_{power} decreased significantly for 2 consecutive days: rest for at most 2 consecutive days, followed by low-intensity training Rest after 9 consecutive days of training
(Nuuttila, Nikander, Polomoshnov, Laukkanen, & Hakkinen, 2017)	32 (32, 0)	19 - 37	$\dot{V}O_{2peak}$: $\sim 53.6 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$	Running	8 weeks training intervention Low-intensity sessions: < individual aerobic threshold Duration: self-selected between 30-90 minutes High-intensity interval sessions: 4x4 min at 90-95% HR_{max} (3-min of recovery); 3x10x30 sec at 95% V_{max} (3 min of recovery between sets) 5 strength-training sessions (including leg press, knee flexion, upper-body exercises, bench step, half squat)	Garmin 920 XT (Garmin Ltd, Schaffhausen, Switzerland)	"Quick recovery test" based on rMSSD (Firstbeat Technologies OY, Jyväskylä, Finland) Supine position 3-min data collection	Training was scheduled in blocks In-between these blocks, HRV data were used to modify training If $HRV <$ baseline: Low-intensity training If $HRV >$ baseline: The next block could be started
(da Silva, Ferraro,	36	18 – 35	Not reported	Running	3 training sessions per	Polar RS800cx (Polar Electro OY, Kempele,	rMSSD measured in the	Baseline: 10 days moving

Adamo, & Machado, 2017)	(0, 36)		week A total of 12 sessions of moderate intensity exercise (6x30 min at 75% v_{max} ; 6x40 min at 75% v_{max}); 12 sessions vigorous intensity exercise at 100% v_{peak} (interval series of total duration comparable to that of the moderate-intensity training) (8 weeks)	Finland)	afternoon/evening	window If rMSSD < baseline: moderately intense exercise If rMSSD >1SD above baseline: High-intensity exercise
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Abbrev.: HRV, heart rate variability; HR_{max} , individual maximal heart rate; HF_{power} , high frequency power; LT1, first lactate threshold; rMSSD, root mean sum of squared differences; SD, standard deviation; v_{max} , maximal running velocity; VO_{2peak} ($ml \cdot kg^{-1} \cdot min^{-1}$), individual maximal systemic oxygen uptake; VT1 2, first and second ventilatory threshold; Wk, week (table adapted from: (Duking, Zinner, Reed, Holmberg, & Sperlich, 2020))

da Silva, D. F., Ferraro, Z. M., Adamo, K. B., & Machado, F. A. (2017). Endurance Running Training Individually-Guided By Hrv In Untrained Women. *J Strength Cond Res*. doi:10.1519/JSC.0000000000002001

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Schmitt, L., Willis, S. J., Fardel, A., Coulmy, N., & Millet, G. P. (2018). Live high-train low guided by daily heart rate variability in elite Nordic-skiers. *Eur J Appl Physiol*, 118(2), 419-428.

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Vesterinen, V., Nummela, A., Heikura, I., Laine, T., Hynynen, E., Botella, J., & Hakkinen, K. (2016). Individual Endurance Training Prescription with Heart Rate Variability. *Med Sci Sports Exerc*, 48(7), 1347-1354. doi:10.1249/mss.0000000000000910

Journal Pre-proof

Table 2: Differences between the HRV-based and predefined endurance training and their effects on performance parameters

	Differences between the HRV-based and predefined training programs	Performance						Comparison of the effects of HRV-based and predefined training
		HRV			Predefined			
Study		PRE	POST	Change (%)	PRE	POST	Change (%)	Result of NHST and/or MBI
(Javaloyes, Sarabia, Lamberts, & Moya-Ramon, 2018)	Training time per week	PPO:	PPO:	PPO:	PPO:	PPO:	PPO:	PPO:
		356.83 ± 39.74	374.28 ± 43.65	5.1 ± 4.5*	346.75 ± 16.73	351.50 ± 17.01	1.4	n.s.
	HRV-based: 9 h 18 min ± 2 h 50 min			Likely beneficial			n.s.	HRV-based:
							Unclear	Possibly beneficial
	Predefined: 8 h 46 min ± 2 h 47 min							
		40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):	40TT (power output):
	Intensity distribution :	243.11 ± 41.73	260.78 ± 44.76	7.3 ± 4.5*	214.42 ± 32.36	223.13 ± 36.15	4.2	n.s.
	(<VT1/ VT1 - VT2/> VT2):			Very likely beneficial			n.s.	HRV-based:
							Possibly beneficial	Likely beneficial
	HRV-based: 66/24/10%							
Predefined: 64/27/9%								
Predefined training involved significantly more time between VT1 & VT2								

(Javaloyes, Sarabia, Lamberts, Plews, & Moya-Ramon, 2019)	Training time per week:	PPO: 395 ± 39	PPO: 423 ± 28	PPO: 7 ± 5*	PPO: 388 ± 42	PPO: 407 ± 51	PPO: 4.9	PPO: n.s.
	HRV-based: 11 h 6 min ± 3 h 4 min			Very likely beneficial			n.s. Possibly beneficial	HRV-based: Unclear
	Predefined: 11 h 22 min ± 3 h 7 min	40TT: 261 ± 28	40TT: 280 ± 39	40TT: 6 ± 6*	40TT: 262 ± 30	40TT: 264 ± 33	40TT: 0.7	40TT: n.s.
	n.s.; (MBI: unclear) Intensity distribution : (<VT1/ VT1 - VT2/ >VT2) HRV-based: 49/39/12% Predefined: 54/33/13% n.s. (MBI: unclear) TRIMP did not differ between groups			Likely beneficial			n.s. Possibly trivial	HRV-based: Unclear
(Schmitt, Willis, Fardel, Coulmy, & Millet, 2018)	HRV-based: 3365 ± 425 a.u.	Roller-ski performance after tapering	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:	Roller-ski performance after tapering:
	Predefined hypoxic: 3481 ± 179 a.u. Daily training by the HRV-based group was adjusted on 3.3 ± 2.3 of the 15 days	n.r.	n.r.	-2.7 ± 3.6*	n.r.	n.r.	-2.5 ± 3.5 n.s.	n.r.
(Vesterinen et al., 2016)	Sessions of moderate- and high-	3000 m (km·h ⁻¹):	3000 m (km·h ⁻¹):	3000 m (%)	3000 m (km·h ⁻¹)	3000 m (km·h ⁻¹)	3000 m (%) 1.1 ± 2.7	3000 m (km·h ⁻¹): 0.42 (small

	intensity: HRV-based: 13.2 ± 6.0* Predefined: 17.7 ± 2.5 Effect size: 0.98 (large)	15.4 ± 1.6	15.7 ± 1.5	2.1 ± 2.0*	15.0 ± 1.6	15.2 ± 1.5	(n.s)	effect)
(Kiviniemi et al., 2010)	Number of moderate-intensity sessions per week: HRV-based: Men: 2.7 ± 0.2* Women: 2.5 ± 0.3* Predefined: Men: 2.1 ± 0.1 Women: 2.2 ± 0.2	Load _{max} (Watt): Men: 270 ± 29 Load _{max} (Watt): Women: 174 ± 28	Load _{max} (Watt): Men: 300 ± 25 Load _{max} (Watt): Women: 189 ± 25	Load _{max} (%): Men: +11.1 Load _{max} (%): Women: +8.6	Load _{max} (Watt): Men: 275 ± 28 Load _{max} (Watt): Women: 179 ± 32	Load _{max} (Watt): Men: 293 ± 35 Load _{max} (Watt): Women: 198 ± 35	Load _{max} (%): Men: 6.5* Load _{max} (%): Women: 10.6*	Load _{max} (Watt): Men: p < 0.1 Load _{max} (Watt): Women: n.s.
(Kiviniemi et al., 2010)	Number of intense sessions of exercise per week: HRV-based: 1.8 ± 0.3* Predefined: 2.8 ± 0.6	Load _{max} (Watt): Women: 177 ± 26	Load _{max} (Watt): Women: 194 ± 23	Load _{max} (%): +9.6	Load _{max} (Watt): Women: 179 ± 32	Load _{max} (Watt): Women: 198 ± 35	Load _{max} (%): Women: 10.6*	Load _{max} (Watt): n.s.
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007)	Number of high-intensity sessions of exercise per week: HRV-based: 3* Predefined: 4	v _{max} (km·h ⁻¹): 15.5 ± 1.0	v _{max} (km·h ⁻¹): 16.4 ± 1.0	v _{max} (%): +5.8 *	v _{max} (km·h ⁻¹): 15.1 ± 1.3	v _{max} (km·h ⁻¹): 15.7 ± 1.2 *	v _{max} (%): 3.9 *	v _{max} (km·h ⁻¹): p < 0.05

(Nuuttila, Nikander, Polomoshnov, Laukkanen, & Hakkinen, 2017)	Number of training sessions per week	3000 m (s): 673 ± 50	3000 m (s): 638 ± 52	3000 m (%): 5.2 *	3000 m (s): 667 ± 47	3000 m (s): 632 ± 41	3000 m (%): 5.2*	3000 m: 0.00 (trivial)
	HRV-based: 6.3 ± 1.4 (3 ± 3% in HR Zone 3)	v_{max} (km·h ⁻¹): 17.6 ± 1.3	v_{max} (km·h ⁻¹): 18.5 ± 1.2	v_{max} (%): +5.1*	v_{max} (km·h ⁻¹): 18.0 ± 1.1	v_{max} (km·h ⁻¹): 18.5 ± 1.2*	v_{max} (%): 2.7 *	v_{max} (km·h ⁻¹): 0.95 (large effect)
	Predefined: 6.1 ± 0.4 (4 ± 3% in HR Zone 3)							
(da Silva, Ferraro, Adamo, & Machado, 2017)	Total relative amount of moderately intense sessions:	v_{max} (km·h ⁻¹): 10.9 ± 1.2	v_{max} (km·h ⁻¹): 11.9 ± 0.9	v_{max} (%): +10.0 ± 7.3*	v_{max} (km·h ⁻¹): 11.0 ± 1.4	v_{max} (km·h ⁻¹): 11.5 ± 1.4	v_{max} (%): 8.2 ± 4.7 %*	v_{max} (km·h ⁻¹): 0.29 (small effect)
	HRV-based: 36.7 ± 5.5 *	t_{lim} at v_{max} (min): 5.1 ± 1.3	t_{lim} at v_{max} (min): 6.1 ± 1.7	t_{lim} at v_{max} (%): +23.6 ± 31.9*	t_{lim} at v_{max} (min): 5.4 ± 1.1	t_{lim} at v_{max} (min): 4.8 ± 1.6	t_{lim} at v_{max} (%): 26.7 ± 31.2 (n.s.)	t_{lim} at v_{max} : 0.93 (moderate effect)
	Predefined: 50 ± 0							
		t5km (min): 36.3 ± 4.5	t5km (min): 29.8 ± 2.4	t5km (%): -11.5 ± 5.6*	t5km (min): 35.5 ± 5.0	t5km (min): 30.5 ± 4.3	t5km (%): -14.0 ± 4.7 *	t5km: 0.67 (moderate effect)

Abbrev.: ES, effect size; 40TT, 40min time trial; 3000m, mean velocity in a 300m run; Load_{max}, maximal load in a cycle ergometer test; v_{max} , maximal running velocity; t_{lim} , time run at maximal velocity; t5km, time to run 5000m; n.r. = not reported; n.s. = not significant; * sign. different (table adapted from: (Duking, Zinner, Reed, Holmberg, & Sperlich, 2020))

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Table 3: Physiological parameters following HRV-based and predefined endurance training

Study	$\dot{V}O_{2peak}$ (ml · kg ⁻¹ · min ⁻¹)							Submaximal parameters						
	HRV		Predefined		Difference between HRV & Predefined			HRV		Predefined		Difference between HRV & Predefined		
	PRE	POST	Change (%)	PRE	POST	Change (%)	Result of NHST and/or MBI	PRE	POST	Change (%)	PRE	POST	Change (%)	Result of NHST and/or MBI
(Javaloyes, Sarabia, Lambert, & Moya-Ramon, 2018)	56.3 ± 7.58	55.8 ± 8.18	n.s. Unlikely beneficial	54.30 (7.81)	52.13 (6.78)	n.s. Very unlikely beneficial	n.s. HRV based: Unlikely beneficial	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):	WVT2 (power output @ VT2):
								275.00 ± 41.46	311.11 ± 37.73	13.9 ± 8.8*	256.25 ± 17.68	281.25 ± 22.16	9.7	n.s.
(Javaloyes, Sarabia, Lambert, & Moya-Ramon, 2018)								WVT1(power output @ VT1):	WVT1(power output @ VT1):	WVT1(power output @ VT1):	WVT1(power output @ VT1):	WVT1(power output @ VT1):	WVT1(power output @ VT1):	WVT1(power output @ VT1):
								191.67 ± 27.95	200.00 ± 25.01	n.s.	175.00 ± 23.15	178.13 ± 28.15	1.7	n.s.

(Javaloyes, Sarabia, Lamberets, Plews, & Moya-Ramon, 2019)	58.9 ± 5.62	61.01 ± 6.01	3 ± 3* Likely Beneficial	58.9 ± 6.23	62.6 ± 6.65	6.2 ± 6.65	n.r.	WVT2: 288 ± 52	WVT2: 349 ± 30	WVT2: 17 ± 15* Very likely beneficial	WVT2: 280 ± 27	WVT2: 323 ± 52	WVT2: 12 ± 12* Very likely beneficial	WVT2: n.s. HRV-based: Unclear
								WVT1: 170 ± 37	WVT1: 234 ± 30	WVT1: 26 ± 18 Very likely beneficial	WVT1: 188 ± 29	WVT1: 190 ± 42	WVT1: 1.1 n.s. Unclear	WVT1: n.s. HRV-based: Unclear
(Schmitt, Willis 2018)	66.7 ± 5.9	69.2 ± 6.5	+3.8 ± 3.1*	63.7 ± 4.4	65.6 ± 4.9	3.0 ± 4.4	n.s.	$\dot{V}O_2VT_2$: 55.3 ± 4.0	$\dot{V}O_2VT_2$: 59.6 ± 5.9	$\dot{V}O_2VT_2$: 6.7 ± 6.1*	$\dot{V}O_2VT_2$: 54.5 ± 5.3	$\dot{V}O_2VT_2$: 57.1 ± 4.7	$\dot{V}O_2VT_2$: 5.0 ± 5.7 n.s.	$\dot{V}O_2VT_2$: n.s.
(Vesterinen et al., 2016)	54.4 ± 6.2	56.4 ± 7.0	+3.7 ± 4.6*	53.0 ± 5.8	55.5 ± 5.8	+5.0 ± 5.2*	0.26 (small negative vs. perfect size)	LT1 (km·h ⁻¹): 11.0 ± 1.2	LT1 (km·h ⁻¹): 11.3 ± 1.2	LT1 (%): +2.8 ± 3.7*	LT1 (km·h ⁻¹): 10.6 ± 1.4	LT1 (km·h ⁻¹): 10.7 ± 1.5	LT1 (%): +1.0 ± 2.9	LT1 (km·h ⁻¹): 0.54 (moderate)
								LT2 (km·h ⁻¹): 13.4 ± 1.4	LT2 (km·h ⁻¹): 13.8 ± 1.3	LT2 (%): +2.6 ± 3.3*	LT2 (km·h ⁻¹): 12.8 ± 1.6	LT2 (km·h ⁻¹): 13.1 ± 1.5	LT2 (%): +1.9 ± 2.2*	LT2 (km·h ⁻¹): 0.25 (small)
(Kiviniemi et al., 2010)	Men							n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
	50 ± 6	54 ± 6	+8.0	50 ± 7	53 ± 7	+6.0*	n.s.	Women						
	36 ± 4	39 ± 3	+8.3	35 ± 5	37 ± 4*	+5.7*	n.s.							

(Kiviniemi et al., 2010)	37 ± 5	40 ± 5	+8.1	35 ± 5	37 ± 4	+5.7	VO _{2peak} : n.s.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
(Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007)	56 ± 4	60 ± 5	+7.1*	54 ± 4	55 ± 3	+1.9	n.s.	VT (km·h ⁻¹):	VT (km·h ⁻¹):	VT (%)	VT (km·h ⁻¹):	VT (km·h ⁻¹):	VT (%)	VT (km·h ⁻¹):
(Nuutila, Nikander, Polomoshnov, Laukkanen, & Hakkinen, 2017)	53 ± 4.2	56.7 ± 3.4	+5.7*	52.2 ± 4.1	56. 4 ± 4.7 *	+8.0*	0.42 (small effect)	LT1 (km·h ⁻¹):	LT1 (km·h ⁻¹):	LT1 (%)	LT1 (km·h ⁻¹):	LT1 (km·h ⁻¹):	LT1 (%)	LT1: 0.32 (small effect)
(da Silva, Ferraro, Adamo, & Machado, 2017)	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	LT2 (km·h ⁻¹):	LT2 (km·h ⁻¹):	LT2 (%)	LT2 (km·h ⁻¹):	LT2 (km·h ⁻¹):	LT2 (%)	LT2: 0.37 (small effect)
	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.

Abbrev.: NHST: Null hypothesis significance testing; LT1, first lactate threshold; LT2, second lactate threshold; MBI, magnitude-based inferences; WVT1, Watt at first ventilatory threshold; WVT2, Watt at second ventilatory threshold; VO_{2peak}, individual maximal systemic oxygen uptake; ES, effect size; 3000 m, time to run 3000 m; n.r. = not reported; n.s. not significant (table adapted from: (Duking, Zinner, Reed, Holmberg, & Sperlich, 2020))

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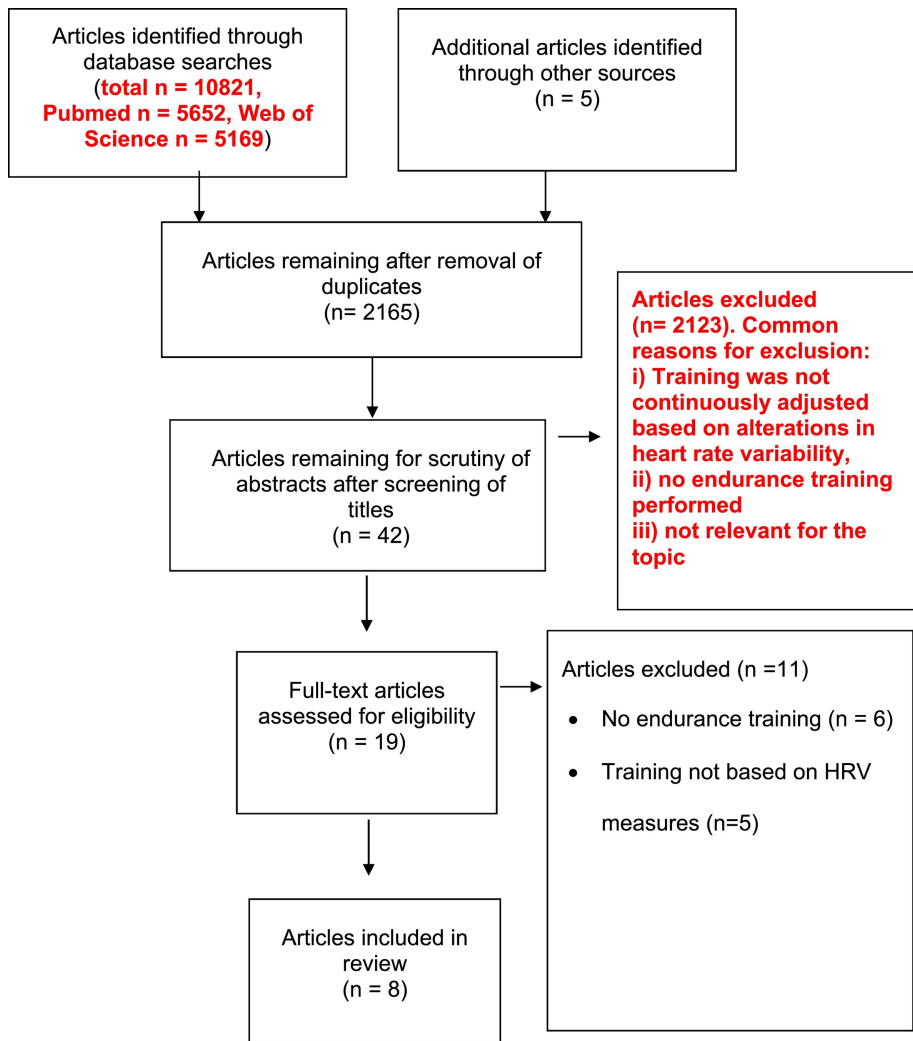


Figure 1

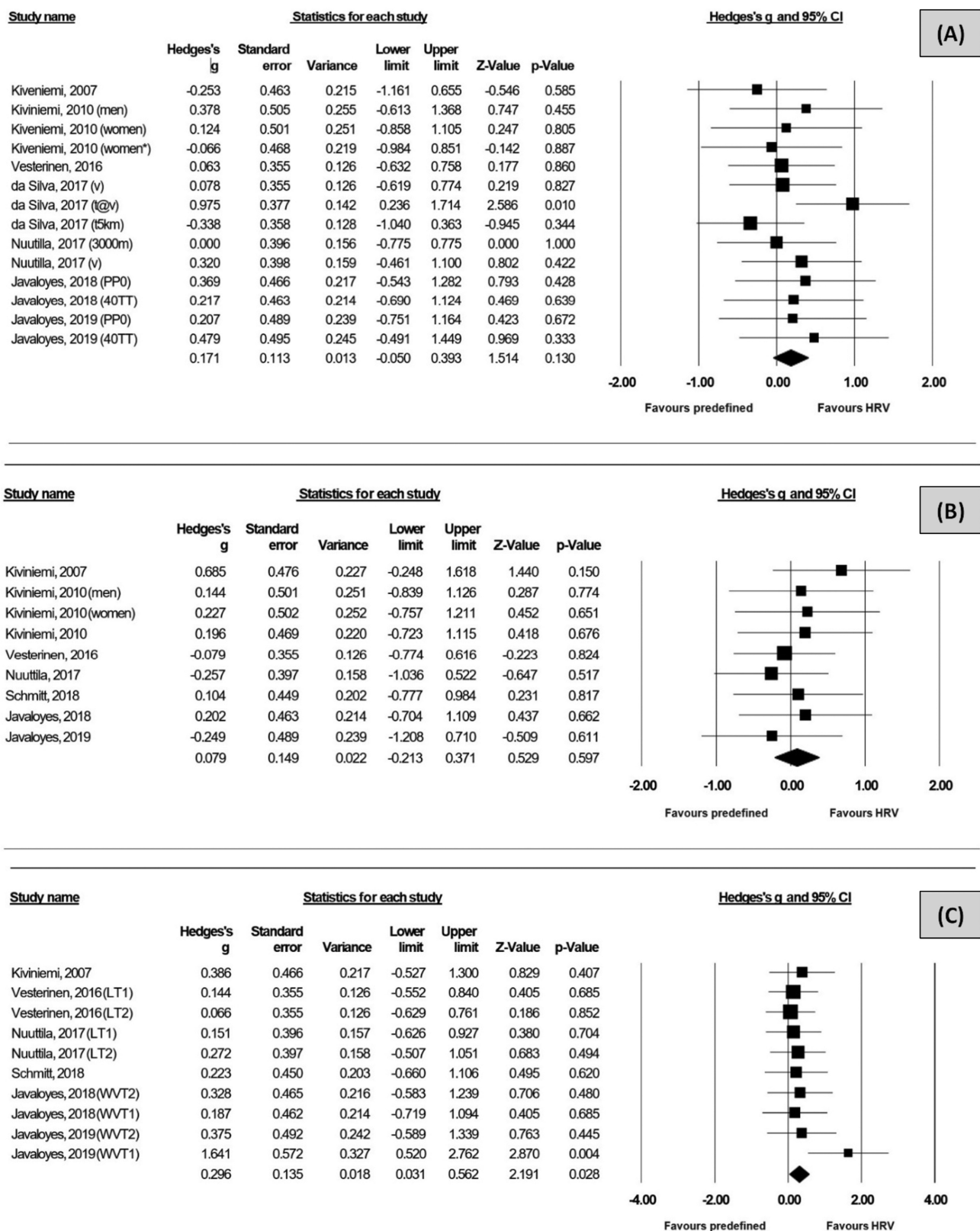


Figure 2

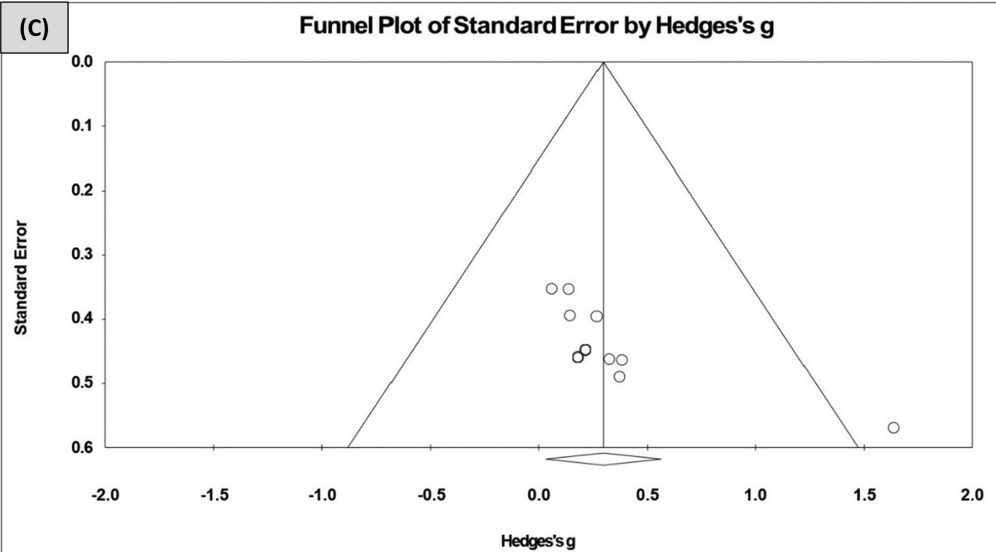
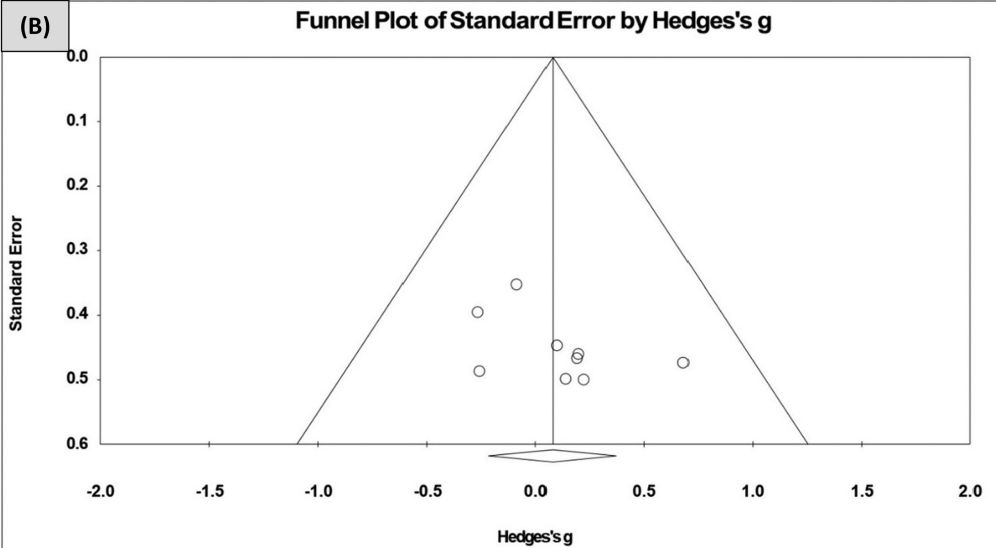
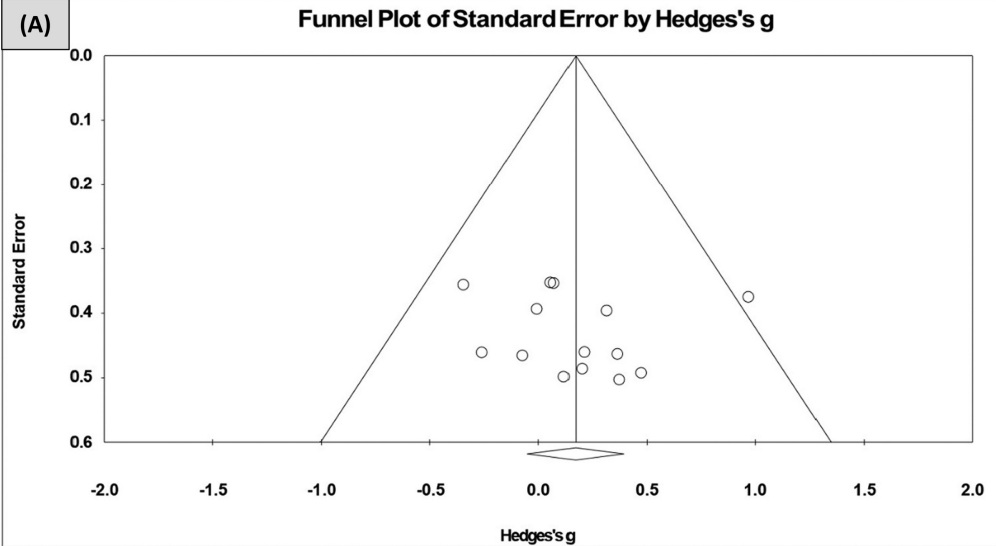


Figure 3